

Docker + Postgres: Troubleshooting & Pro Setup Notes

Core Mental Model

- Containers have their own network namespace.
- 'localhost' inside a container refers to the container itself.
- Services communicate via Docker service names (e.g. postgres, redis).

Why the Error Happened

- The API connected correctly using PGHOST=postgres.
- node-pg-migrate defaulted to localhost:5432.
- Local Postgres conflicted with container networking.

Correct Way to Run Migrations

- Run migrations inside the API container.
- Use docker compose exec api npm run migrate:up.
- Ensure DATABASE_URL points to postgres service, not localhost.

Environment Variable Rules

- Postgres image uses POSTGRES_* only at container init.
- pg client uses PG* variables or DATABASE_URL.
- Changing env vars requires container rebuild if DB already exists.

Production-Style Setup

- Do not expose database ports unless absolutely necessary.
- Use healthchecks to control startup order.
- One execution context = one source of truth.

Key Takeaways

- Infrastructure bugs feel worse because they span systems.
- Docker issues are usually networking or env scope problems.
- This setup now mirrors real production environments.